AI Workforce Odyssey

Agent-Based Simulation Memo | CO (Coding) 1

1. Model Overview

This simulation models workforce transitions in an AI-enabled economy. Thirty workers with varying task profiles, seniority levels, and behavioral traits face accelerating automation pressure over 75 time steps. The model tracks how workers move through five states — **Employed, At-Risk, In-Training, Unemployed, and Recovered** — using seven explicit transition rules. The core question is whether equal access to training produces equal recovery. Two scenarios are compared using the same model structure: **Tech-Driven Automation** (rapid AI adoption, limited support) and **Human-Centric Support** (selective adoption, stronger training infrastructure and augmentation pathways).

2. Worker Design

Workers are distributed across four task types reflecting a spectrum of AI exposure:

Task Type	Count	Routine-Share	AI Risk	Examples
Routine-Cognitive	10	70–90%	High	Data entry, bookkeeping
Non-Routine Cognitive	8	15–35%	Low	Analysis, design
Relationship-Based	6	40–55%	Moderate	Sales, partnerships
Non-Routine Manual	6	10–30%	Low	Care work, trades

Each worker carries **two behavioral traits: adaptability** (governing training entry, 0–100) and **resilience** (governing susceptibility to disengagement, 0–100), plus a **seniority level** (Junior, Mid, or Senior) where juniors face higher disruption risk. **Disruption risk** is computed from routine-share, automation pressure, and seniority, normalized to 0–1 to produce staggered disruption across task types.

3. Emergent Mechanism

The model includes seven programmed **transition rules** governing individual worker behavior **(threshold-based disruption, probabilistic training entry, timeout-based unemployment, training completion, re-entry with scarring, disgruntled contagion, and recovered re-risk)**, plus **capped training seats, peer-boost from nearby recovered workers, and a tunable scarring rate.** Full rule specifications are documented in the design spec and .nlogo file. The key distinctions between these programmed rules and the system-level patterns that emerged from their interaction:

Adaptation inequality: Workers with identical training access but different adaptability, timing, and spatial positions produced divergent outcomes. Some recovered quickly; others became permanently trapped. This was not programmed — it emerged from individual heterogeneity interacting with shared resource constraints.

Unemployment traps from scarring: The scarring mechanism created a feedback loop: longer unemployment reduced adaptability, which reduced re-entry probability, which extended unemployment further. In Tech-Driven, this produced a persistent tail of unemployed workers, particularly routine-cognitive juniors disrupted earliest.

Recovery and stagnation clusters: Peer influence created spatial pockets of recovery near early-recovered workers and stagnation in areas without peer support. These clusters were not placed by design — they emerged from fixed random placement and the peer-boost mechanic.

Queue amplification under stress: Training seats were calibrated as a secondary constraint. In Tech-Driven, concentrated disruption waves temporarily exhausted capacity (1 seat available at tick 75), amplifying the gap between high- and low-adaptability workers.

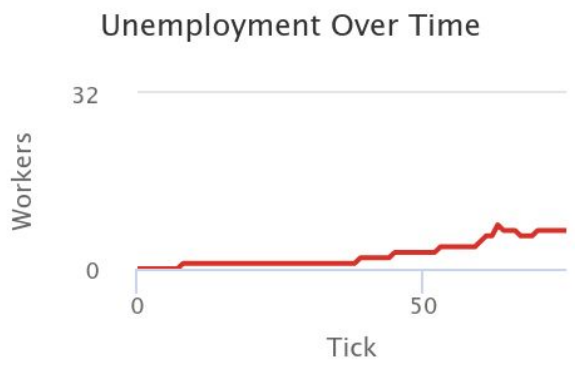
4. Scenario Results

Both scenarios used identical code with different parameter values. Results at tick 75:

Metric	Tech-Driven	Human-Centric
Employed	6	8
In Training	7	12
Unemployed	9	1
Recovered	5	6
% Augmented	6.7%	10%
Avg Time Unemployed	10.6 ticks	5.3 ticks
Training Seats Available	1	8

Figure 1: Unemployment Over Time

Tech-Driven



Human-Centric

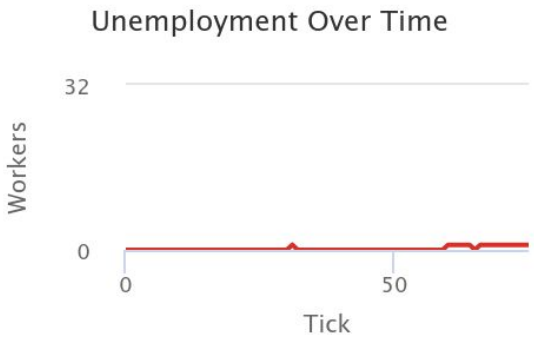
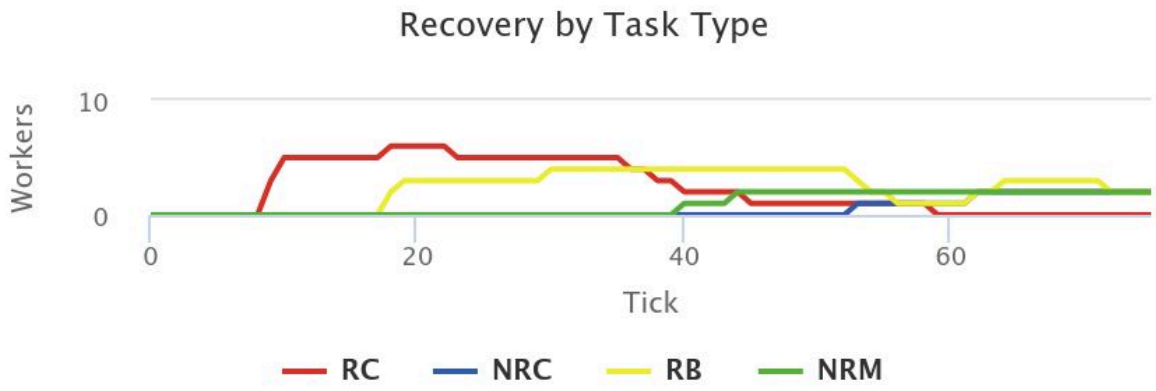
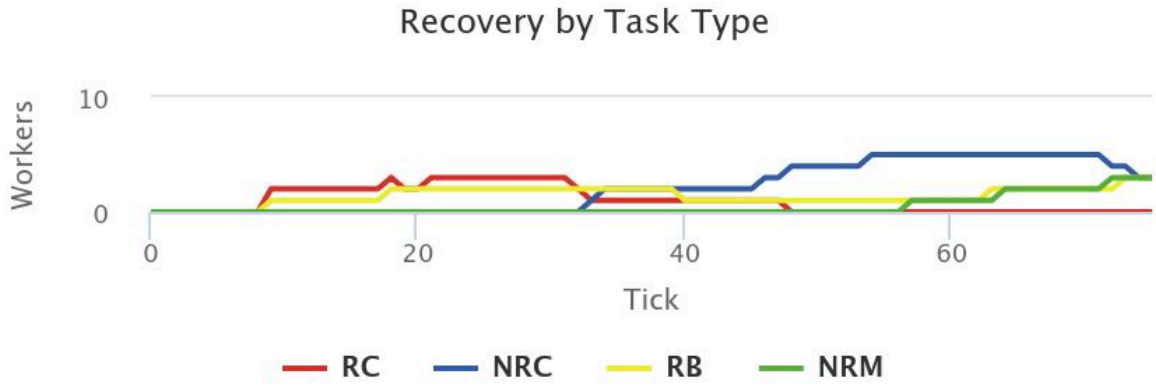


Figure 2: Recovery by Task Type

Tech-Driven



Human-Centric



Interpretation

The most striking difference is unemployment: 9 workers trapped in Tech-Driven versus just 1 in Human-Centric. This is not simply because Human-Centric has more training seats — seats were non-binding (8 still available at tick 75). **The difference arises from the interaction of slower disruption timing, lower scarring rate, and wider peer influence radius.**

As Figure 1 shows, **Tech-Driven unemployment** rises steadily after tick 50 and stays elevated, reflecting the **scarring feedback loop** where early-displaced workers lose adaptability faster than they can regain access to training. **Human-Centric unemployment** remains near zero — not because workers avoid disruption, but because **slower pressure and stronger support** allow workers to move through the pipeline before scarring accumulates.

The training bottleneck behaved as designed: secondary, not dominant. In Human-Centric, it never bound. In Tech-Driven, it bound only during peak disruption waves. This confirms the model's core insight: **the primary driver of divergence is unequal adaptation capacity, but institutional constraints can amplify or dampen that divergence.**

As Figure 2 illustrates, **routine-cognitive workers were consistently the most affected group in both scenarios.** They recover first but lose that recovery under sustained automation pressure, while non-routine workers recover later but more durably — a pattern that emerged from the **interaction of accelerating pressure and task-type differences.** Human-Centric support doesn't just increase recovery overall; it equalizes outcomes across task types.

5. Reflection

This model demonstrates that workforce disruption from AI is not a single-variable problem. The same workers, facing the same technology, can experience fundamentally different outcomes depending on the **speed of adoption, the availability of support, and the local dynamics of their peers.** The central finding — **that equal training access does not produce equal recovery** — emerges naturally from the **interaction of individual heterogeneity, scarring feedback, and peer influence**, rather than being programmed directly.

The model has limitations: it includes only workers as agents (firms and policy are parameters only), uses a fixed population of 30 with no entry or exit, relies on spatial proximity as an abstraction of professional networks, and uses modeling assumptions not calibrated to empirical data. Despite these constraints, the model illustrates how simple individual-level rules generate complex system-level patterns, and how institutional context determines whether technological disruption produces widespread recovery or persistent inequality.

Appendix: Technical Details & AI Use

Transition Rules (Full Specification)

Seven transition rules govern individual worker behavior: (1) Threshold rule: Employed → At-Risk when disruption risk exceeds a scenario-specific threshold. (2) Probabilistic training entry: At-Risk → In-Training based on adaptability + peer boost – disgruntled penalty, if seats available. (3) Timeout: At-Risk → Unemployed after a set number of ticks without entering training. (4) Training completion: In-Training → Recovered after a set duration, with augmented vs re-employed determined by probability. (5) Re-entry with scarring: Unemployed → In-Training, but adaptability erodes by scarring-rate per tick spent unemployed, making re-entry progressively harder. (6) Disgruntled contagion: At-risk workers with resilience below 40 become disgruntled after 3 ticks; nearby disgruntled workers erode others' resilience by 2 per tick. (7) Recovered re-risk: Recovered workers return to At-Risk if disruption risk exceeds threshold + 0.15 buffer.

Additionally, training seats are capped as a parameter (8 Tech-Driven / 20 Human-Centric), peer-boost provides +5 adaptability per nearby recovered or in-training worker (capped at 20), and scarring reduces adaptability by a tunable rate (0.5/tick Tech-Driven, 0.3/tick Human-Centric).

Scenario Parameters

Parameter	Tech-Driven	Human-Centric
Base pressure	10	5
Acceleration rate	0.4 /tick	0.20 /tick
Disruption threshold	0.50	0.55
Training seats	8	20
Training duration	8 ticks	5 ticks
Augmentation probability	30%	55%
At-risk timeout	6 ticks	10 ticks
Disgruntled timeout	8 ticks	12 ticks
Peer influence radius	2 patches	4 patches
Scarring rate	0.5 /tick	0.3 /tick

Iterative Design & Parameter Tuning

The model design went through multiple revisions (v1 → v2 → v3 → v3.1 → v3.2), incorporating external feedback at each stage. Key changes included simplifying from 7 states to 5 (converting Augmented and Disgruntled from states to attributes), normalizing the disruption-risk formula to prevent numerical explosion, and reframing the training bottleneck as secondary rather than dominant.

Initial parameter values produced a Tech-Driven scenario that was too harsh (scarring rate of 1.0/tick created permanent unemployment traps, 0% augmented workers) and a Human-Centric scenario that was too gentle (threshold of 0.60 left most workers undisrupted). After test runs, parameters were tuned: scarring rate reduced to 0.5/0.3, Tech-Driven threshold raised to 0.50, Human-Centric threshold lowered to 0.55 and acceleration increased to 0.20, augmentation probability raised to 30%.